

# ABD ELAZIZ HAFALLAH

## Software Engineer | Generative AI & Agentic Systems | Full Stack

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### SUMMARY

Software engineer with 2+ years building production backend and full-stack systems for Algeria's largest classifieds marketplace, serving millions of users and requests (microservices, REST/GraphQL APIs, RabbitMQ, Docker, Kubernetes, Keycloak). Now building reliable agentic AI systems with human-in-the-loop controls, evaluation harnesses and observability. MSc Software Engineering student at the University of Innsbruck.

### TECHNICAL SKILLS

**Agentic AI & GenAI:** LangGraph, multi-agent orchestration, tool calling, plan/verify loops, human-in-the-loop, RAG, hybrid search (BM25 + vector), reranking, embeddings, structured outputs, evaluation harnesses, prompt-injection defence, LLM APIs (OpenAI, Anthropic, Azure OpenAI)

**Languages:** Python, TypeScript, JavaScript, PHP, SQL, C/C++

**Backend & Frontend:** FastAPI, NestJS, Node.js, Laravel, REST, GraphQL, Microservices, RabbitMQ, React, Vue.js 3

**Data:** PostgreSQL, pgvector, Redis, MySQL, MongoDB, SQLAlchemy, Prisma ORM

**DevOps & Security:** Docker, Kubernetes, GitLab CI/CD, GitHub Actions, Linux, Git, Keycloak, OAuth 2.0 / OpenID Connect, JWT, RBAC

**Observability & Tools:** Sentry, OpenObserve, OpenTelemetry, Prometheus, Grafana, Jira, Agile/Scrum

### AI ENGINEERING PROJECTS

#### RootCause - Multi-Agent SRE Assistant

github.com/aziz-05/rootcause-agent

Python, LangGraph, FastAPI, PostgreSQL/pgvector, Redis, Docker, Kubernetes, OpenTelemetry

- Diagnoses infrastructure incidents through a plan → execute → verify agent loop against a demo microservices stack with genuinely injected faults (container kills, network latency, connection-pool exhaustion).
- Write actions require recorded human approval enforced in code and database; fixed tool registry and Kubernetes RBAC limit remediation scope.
- 23-check invariant gate on every pull request, including adversarial prompt-injection cases; 53 tests.

#### Souk Copilot - Production RAG & Analytics Platform

github.com/aziz-05/souk-copilot

Python, FastAPI, PostgreSQL/pgvector, Redis, Docker, OpenAI / Azure OpenAI

- Routes policy and analytics questions to a retrieval or text-to-SQL path, refusing rather than guessing below a relevance floor.
- Hybrid retrieval (BM25 + vector, reciprocal rank fusion, reranking, section-aware chunking) with citations bound to source chunks.
- CI-gating evaluation on recall@k, MRR, refusal correctness and groundedness; AST-validated read-only SQL sandbox; JWT with row-level isolation; 103 tests.

### PROFESSIONAL EXPERIENCE

#### Full Stack Web Developer | SARL Ouedkniss, Algiers, Algeria

Nov 2023 - Present

- Built and maintained microservices and REST/GraphQL APIs (Laravel, NestJS, Node.js) behind React and Vue front-ends for a platform serving millions of users and requests, keeping core listing and search features fast and stable under heavy traffic.
- Worked directly with production databases (PostgreSQL, MySQL) and live data at scale, optimising slow queries and applying data changes safely, which improved response times while preserving data integrity.
- Implemented asynchronous, event-driven processing with RabbitMQ to decouple services and move heavy work off the request path, improving API responsiveness during traffic peaks.
- Secured user and service access with Keycloak (OAuth 2.0 / OpenID Connect) and role-based access control, centralising authentication across microservices.
- Containerised services with Docker and deployed them to Kubernetes through GitLab CI/CD pipelines with automated tests, making releases faster, repeatable and lower-risk.
- Integrated Sentry error tracking and OpenObserve logs and metrics, shortening the time to detect, diagnose and fix production incidents.
- Used AI coding agents in daily development, testing and code review, and delivered sprint work tracked in Jira, speeding up feature delivery.

#### Network Engineer | Freelance, Algeria

Mar 2020 - Nov 2021

- Built and maintained LAN/WAN and VPN infrastructure for small businesses, including patching and security configuration.

### EDUCATION

#### MSc Software Engineering (in progress) | University of Innsbruck, Austria

Oct 2026 - Present

Coursework: Machine Learning, Parallel Programming, Logic.

#### MSc Networking & Distributed Systems | University of Constantine 2 - Abdelhamid Mehri, Algeria

2019 - 2021

#### BSc Computer Science | University of Constantine 2 - Abdelhamid Mehri, Algeria

2016 - 2019

Coursework: distributed systems, microservices, network security, AI & deep learning. Projects: EMODEC speech-emotion recognition API; QoS-AODV VANET routing protocol (NS-2, C++).

### LANGUAGES

Arabic (Native) | English (B2) | French (B2) | German (in progress)